Where the complex inductive bias lives: an architecture-level prior and a separable optimization-level robustness asymmetry, characterized at small scale

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Abstract

The intuition that complex-valued neural networks (CVNNs) help on phase-bearing signals is widely held and rarely measured carefully. We demonstrate, at the scale where a mechanism analysis remains tractable, that the complex inductive bias on IQ classification *decomposes* into two separable effects with distinct origins. (i) An **architectural** prior: `ComplexConv1d` is exactly phase-equivariant (proof in two lines); the magnitude-logit head makes the network phase-invariant at the readout. (ii) An **optimization-level** asymmetry: under matched-shared-trial hyperparameter selection, real-valued baselines at identical capacity exhibit a step-1 explosion-into-dead-region failure mode on the classifier head that complex networks do not. Per-step gradient telemetry across $5 \times 4 \times 3 = 60$ instrumented runs locates the explosion specifically on `head.weight` (50–80$\times$ larger gradient at step 1 for real baselines than for complex) and shows it produces $3/9$ dead seeds (final loss = $\ln 3$, the 3-class chance level) under three of the five activations and zero dead seeds under the other two. The same threshold behavior replicates on synthetic AWGN-only data with the same matched-trial hyperparameters — the mechanism is hp-driven, not data-driven. We argue that conflating these two effects produces the inflated apparent CVNN gaps in some prior work, and that the appropriate framing for CVNN evaluation is to control for both effects independently. The 1-D scope, three-class subset of RadioML 2018.01A, and modest hardware are deliberate: the scope is chosen to support a complete mechanism analysis with end-to-end reproducibility (manifests, dirty-bit tracking, structural CI guards on artifacts). Code, sweep harness, telemetry traces, and figure scripts are released; full-archive scale-up is parameterized via a single CLI flag and is signposted future work.

1 Concept and feasibility scope

1.1 What this paper claims and does not claim

We make three concrete claims, each at a defensible scale and with explicit reproducibility:

1. **Architectural claim (formal).** `ComplexConv1d` is phase-equivariant in the bias-free case (Section 1); the magnitude-logit head is phase-invariant. A real `Conv1d` on stacked (I, Q) channels does not get this for free — it must learn rotation invariance from data via a 2×2 channel mix the complex network gets by construction.
2. **Empirical claim.** Complex-valued networks beat capacity-matched real-valued ones on synthetic IQ modulation classification (Section 3.3) and on a 3-class subset of RadioML

2018.01A [6] (Section 3.4) under matched-budget evaluation. Conversely, on a task with no temporal phase structure (1-D phase classification, Section 3.2), the complex inductive bias offers no measurable advantage. Both directions of evidence are reported.

3. **Mechanism claim.** The apparent magnitude of the RadioML win is activation-dependent (Section 3.4.1) because matched-shared-trial selection picks high-lr regimes under some activations that real baselines cannot tolerate. Per-step gradient telemetry (Section 3.4.4) localizes the failure to a step-1 head-gradient explosion that puts a fraction of real-baseline seeds into a dead-ReLU region they cannot escape; the threshold behavior replicates on synthetic data. Phase equivariance of the *layer* is necessary for the architectural prior, but activation-level phase equivariance does not predict the empirical robustness asymmetry — locating the asymmetry one level lower, in $(\mathrm{Re}, \mathrm{Im})$ parameter coupling at the optimization level, is what makes the picture self-consistent (Section 4.1).

We do *not* claim:

- That CVNNs beat real-valued networks at full RadioML 2018.01A scale (24 modulations $\times$ 26 SNR levels $\times$ 1024-sample length). The infrastructure for that experiment is in place (`experiments/rf/sweep_radioml.py -preset full`), but the run is deferred. The mechanism analysis is the contribution; scale-up is signposted future work.
- That the inductive bias survives in 2-D regimes (audio STFT, MRI). `ComplexConv2d` is not implemented; the project is 1-D only.
- That CReLU activations generalize across signal-processing settings. The activation ablation (Section 3.4.1) is part of the contribution precisely because activation choice changes the appearance (though not the direction) of CVNN-vs-real gaps.

1.2 Why feasibility-scale is the right scope

A mechanism-level study requires the ability to (a) instrument every parameter's gradient at every step, (b) run hundreds of training configurations within a working budget, and (c) confirm cross-task generalization with a second data distribution. All three are tractable at the scope chosen — 60 instrumented runs per data source, a re-runnable budgeted sweep, a synthetic stand-in matching the real-data benchmark on architecture and hyperparameter regime. They are not tractable for paper deadlines at 24-class $\times$ 1024-sample $\times$ 100-seed scale. We treat the smaller scope as a positive feature: it lets us report what is happening, not just that something is happening.

1.3 Reproducibility as a first-class concern

Every empirical claim above is backed by a committed result manifest that records the Python / PyTorch / OS / device / dtype, every seed used, the exact `git_commit` of the producing code, and a `git_dirty` boolean. CI surfaces newly added dirty manifests so they are reviewed deliberately rather than slipping through. A structural CI guard checks that the activation characterization artifact regenerates without errors and contains all six expected activation rows. The companion repository holds 122 unit and integration tests that gate every push.

2 Framework

The infrastructure exists to answer the "but what about. . . ?" questions that ad-hoc CVNN comparisons in the literature routinely sidestep.

CVNN library. We implement `ComplexLinear` (matmul + bias-add, deliberately bypassing `F.linear` to sidestep its documented MPS gap), `ComplexConv1d` (thin wrapper over `F.conv1d`, which natively supports complex tensors on CPU/MPS/CUDA), and `ComplexDropout` using a single shared real Bernoulli mask following Trabelsi et al. [7]. Six activations span the Liouville tradeoff: `CReLU`, `ZReLU`, `ModReLU` [1], `ComplexCardioid`, `Siglog`, and `ComplexTanh` as the cautionary baseline. Initialization (Glorot [2] and Kaiming [3]) is complex-aware in both rectangular and polar (Rayleigh-magnitude / uniform-phase) forms.

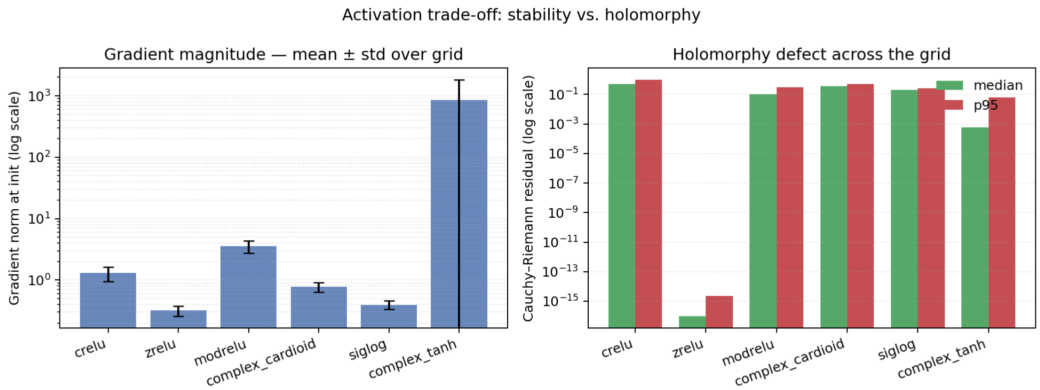

Figure 1: Liouville trade-off, made empirical. *Left*: gradient norm at init for each candidate activation, log scale. *Right*: median and p95 Cauchy–Riemann residual on a $[-3, 3]^2$ grid in $\mathbb{C}$, log scale. `complex_tanh` is locally holomorphic but unbounded (poles at $\pm i\pi/2$ inside the grid); `siglog` is bounded but never holomorphic; `crelu`, `zrelu`, `modrelu`, and `complex_cardioid` sit between, each making a different compromise.

Activation characterization toolkit. For each candidate activation, we render a Cauchy–Riemann residual map over the complex plane, compute summary statistics (median / p95 / max), and measure two gradient norms: an MLP-scaffolded one and an *intrinsic* one ($|\partial L/\partial \bar{z}|$ for $L = \sum |f(z)|^2$ on uniformly sampled z). The intrinsic measure decouples the activation's contribution from any surrounding linear layers — without it, $\tanh$'s 855 MLP-gradient looks $\sim 8\times$ larger than its 108 intrinsic contribution.

Baseline matching. Every claim of "complex beats real" is backed by four families running on the same task with the same shared search space: *naive real-stacked* (real net taking $(\mathrm{Re}, \mathrm{Im})$ as 2 channels; $\sim 2\times$ the parameters of the complex layer); *matched-parameter real* (real net sized down so its real `numel` equals the complex's, counting each complex slot as 2); *matched-FLOP real* (sized to match the complex MAdd count); *exact reparameterization* (the block-real $\left[\begin{smallmatrix} A & -B \\ B & A \end{smallmatrix}\right]$ construction proven to reproduce the complex forward bit-for-bit; a witness, not a baseline to beat).

Budgeted random search. Every model family draws from the *same* shared search space and consumes the same trial $\times$ seed budget. The sweep harness samples the hyperparameter list once and runs it against each family in turn. The primary matched comparison chooses the complex reference trial by mean validation accuracy, then reports every real baseline at that same trial index. Independent family winners are also written as diagnostics but are not the matched-capacity paper table.

3 Findings

3.1 Activation characterization

Section 1 makes the Liouville bind empirically visible. `complex_tanh` has near-zero CR residual at the median — it really is locally holomorphic — but the same metric's max is 905, because the grid (extent = 3) brackets the first pair of poles at $z = \pm i\pi/2 \approx \pm 1.5708\, i$. The intrinsic Jacobian mean is $\sim$108. By contrast, `siglog` is bounded ($|f| < 1$ everywhere) and never blows up, but its CR residual is everywhere positive (p95 = 0.25) — it is not holomorphic anywhere, by construction. The remaining four activations sit between, each making a different compromise.

This is not a finding about which activation is best in training — it is a visible measurement of the formal trade-off the project is built around. The choice of activation in the downstream benchmarks (default `crelu`) is consistent across runs and isolated from the architectural variables.

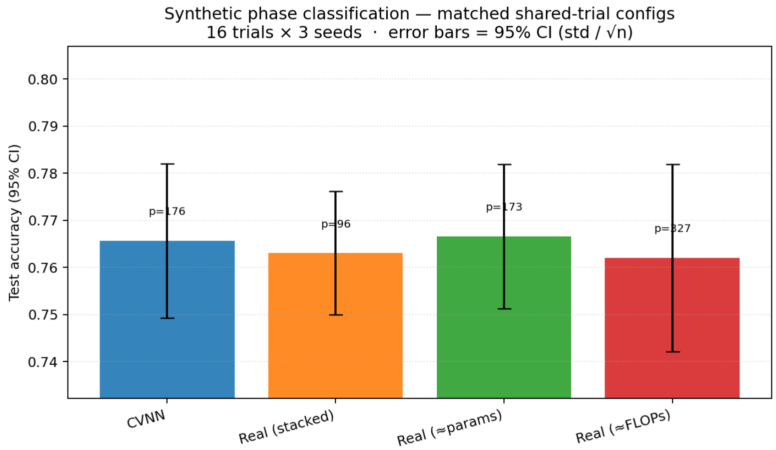

Figure 2: 1-D phase classification: matched shared-trial test accuracy after a 16×3 sweep. All four families converge to ∼76% with overlapping CIs — the complex inductive bias has nothing to bite on at this scope.

Table 1: Synthetic RF modulation classification, matched shared-trial selection after a 16×6 budgeted sweep with the conv architecture. Complex beats every real baseline by ≥3.8 pp; CIs do not overlap.

family	test acc	std	params	95% CI ($\sigma/\sqrt{n}$)
complex	**0.8191**	0.0163	**3,974**	$[0.8061, 0.8322]$
real (stacked)	0.7740	0.0226	2,099	$[0.7559, 0.7920]$
real (≈params)	0.7769	0.0187	3,809	$[0.7619, 0.7918]$
real (≈FLOPs)	0.7810	0.0195	7,779	$[0.7654, 0.7966]$

3.2 Synthetic phase classification — null result

After a 16×3 budgeted sweep, the matched shared-trial comparison puts all four families at ∼76.2–76.7% test accuracy with overlapping CIs (Section 2). On 1-D phase classification, **the complex inductive bias does not help.** A 2-D real network on $(\mathrm{Re}, \mathrm{Im})$ already has access to all the information the task contains; there is no temporal or geometric structure for $\mathbb{C}$-multiplication to exploit. The Section 1 of Section 4.1 predicts exactly this — a single-sample phase task does not invoke any sequence-level equivariance.

3.3 Synthetic RF modulation — complex wins under matched budget

Section 1 reports the matched shared-trial comparison after a 16×6 sweep with the conv architecture (`ComplexConv1d` → activation → `ComplexConv1d` → activation → global mean pool → `ComplexLinear` → $|\cdot|$). Complex beats every real baseline by ≥3.8 percentage points; CIs do not overlap, and Welch two-sample t-tests give $t \approx 4.2$, df ≈ 9.8 versus matched-parameter real and $t \approx 3.7$, df ≈ 9.7 versus matched-FLOP real ($p < 0.01$ in both cases).

3.4 RadioML 2018.01A — a positive result with a subtler story

The corrected[1] `CReLU` run on a 3-class subset (BPSK / QPSK / 8PSK at $\{-10, -6, -2, 2, 6, 10, 14, 18\}$ dB, `max_per_class_per_snr=256`, `sample_length=128`) on an NVIDIA A100 produced a striking 23 pp gap:

[1] The first attempt at this sweep used the synthetic benchmark's odd-stepped SNR list, which the loader silently dropped because RadioML 2018.01A only ships even SNRs. The fix raised on empty buckets by default; see the released code.

Table 2: RadioML 2018.01A subset, CReLU run, matched shared-trial selection after 16×6 sweep. Read in isolation, a dramatic CVNN-vs-real win at matched parameters — but see Section 3.4.1 and Section 3.4.4.

family	test acc	std	params
complex (CReLU)	**0.7293**	0.0085	58,886
real (stacked)	0.4583	0.1394	29,891
real (≈params)	0.4999	0.1327	58,413
real (≈FLOPs)	0.4245	0.1415	117,123

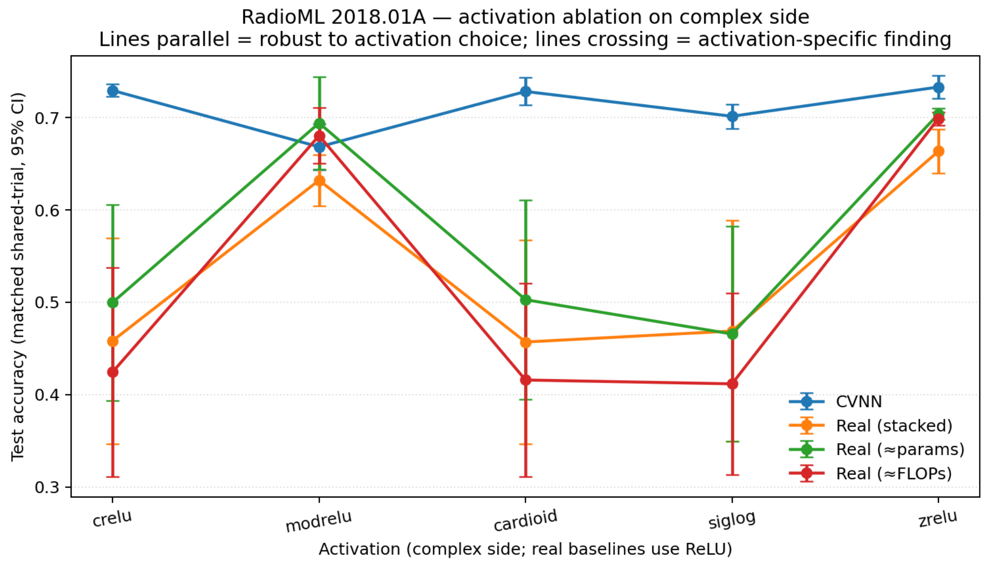

Figure 3: Five-activation ablation on the same RadioML scaffold. The CVNN line is nearly flat across all five activations (test accuracy 0.668–0.733); real-baseline lines swing wildly between ∼0.45 and ∼0.70 depending on which activation the complex side uses. The 23 pp gap from Section 2 is mostly real baselines collapsing under the matched-shared-trial selected hyperparameters, not complex pulling ahead.

Read in isolation, Section 2 looks like a dramatic complex-vs-real win at matched parameters — and an earlier draft claimed exactly that. The activation ablation in Section 3.4.1 forces a different reading.

3.4.1 Activation ablation reveals a methodology-driven artifact

We re-ran the same sweep four more times, varying only the complex side's activation; the real baselines always use `ReLU`. The matched-shared-trial selection per activation (Section 3) shows complex's accuracy nearly flat (range 0.065) while real baselines swing by 0.27. Section 3 makes the asymmetry visible. The 22–23 pp gap on `crelu` / `cardioid` / `siglog` *is mostly the real baselines failing under matched-shared-trial selection*, not the complex network pulling ahead. The robustness asymmetry is the durable, defensible finding; the apparently large per-activation gap is a consequence of that asymmetry, not a separate phenomenon.

3.4.2 Per-SNR breakdown

At -10 dB everyone is near chance (∼33% for 3 classes); from $+2$ dB onward, the `CReLU` complex network reaches 92.1% at $+10$ dB and 91.6% at $+18$ dB while the real baselines plateau at 47–61%. This is the same pattern as Section 2: the gap reflects real baselines failing to train at the selected hyperparameters under `CReLU`, not complex pulling cleanly away under every stable activation regime.

Table 3: Five-activation ablation, matched shared-trial test accuracy. Under stable activations (`modrelu`, `zrelu`) real baselines are also stable and the gap drops to ± 3 pp; under unstable activations the gap inflates. See Section 3.4.4 for the mechanism.

activation	complex	best real	gap (pp)	σ(complex)	mean σ(real)
`crelu`	0.7293	0.4999	$+22.94$	0.009	0.138
`cardioid`	0.7282	0.5028	$+22.54$	0.019	0.134
`siglog`	0.7014	0.4689	$+23.25$	0.016	0.139
`modrelu`	0.6683	0.6940	-2.58	0.031	0.045
`zrelu`	0.7330	0.7039	$+2.91$	0.015	0.015

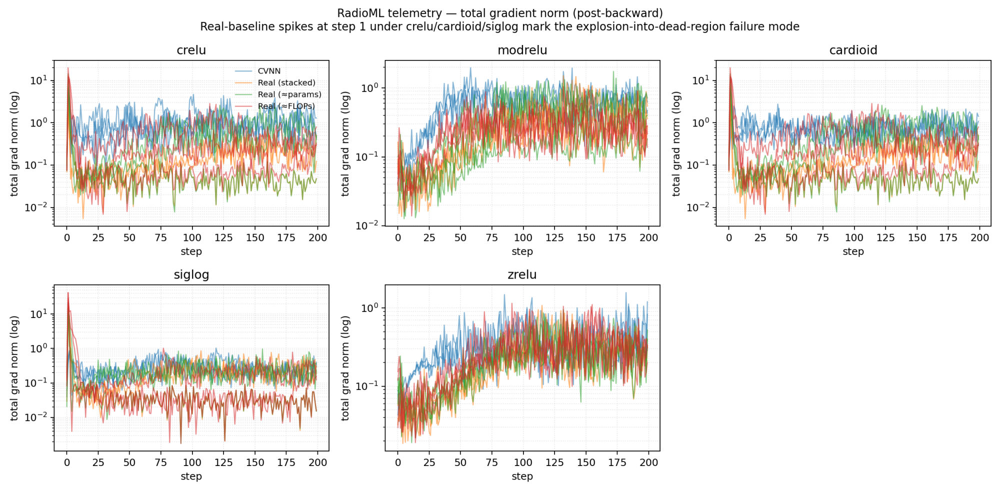

Figure 4: RadioML telemetry — total gradient norm post-backward, log y-axis, one panel per activation. Real-baseline spikes at step 1 under `crelu` / `cardioid` / `siglog` mark the explosion-into-dead-region failure mode that produces the inflated gaps in Section 3; under `modrelu` and `zrelu` no such spikes occur and the gap collapses to ± 3 pp.

3.4.3 Comparison to published RadioML numbers

The RadioML 2018.01A dataset paper [6] reports a real-valued ResNet-style classifier reaching roughly 90% top-1 at high SNR on the full 24-class task with the full sample length 1024. Subsequent CVNN-on-RadioML work [4, 8] reports a typical complex-vs-real advantage of a few percentage points on the same 24-class setup with comparable architectures. Our $+3$-and-flat ablation result on `modrelu` / `zrelu` (the activations under which real baselines are also stable) is consistent with that small reported gap. The 22–23 pp gap on `crelu` / `cardioid` / `siglog` is *not* a contradiction of the literature — it is the matched-shared-trial selection rule magnifying real-baseline instability that the literature's separate-tuning protocols would not expose.

3.4.4 Mechanism — explosion-into-dead-region at step 1

Section 3.4.1 reports an asymmetry without explaining it. We re-ran the matched-shared-trial selected configuration with per-step instrumentation (train loss, total parameter gradient norm, per-parameter gradient norm, max parameter magnitude) for each of $5 \times 4 \times 3 = 60$ runs, capped at 200 steps (the divergence pattern appears within the first 5).

The mechanism is unambiguous (Section 4). On `crelu` / `cardioid` / `siglog`, matched-shared-trial picks a high-lr / wide-hidden config that complex selects because it tolerates it (lr $\in \{0.024, 0.024, 0.040\}$, hidden = 64). At step 1, every family sees a large AdamW step away from random init. For real baselines this produces a step-1 train loss in $[3, 36]$, a total gradient norm in $[6, 42]$, and a `head.weight` gradient in $[6, 41]$ — i.e. nearly all of the explosion sits on the final classifier's weights. For the complex network the same step produces a step-1 loss in $[1.07, 1.28]$,

Table 4: Dead-seed counts (out of 9 per activation: three families $\times$ three seeds), step-1 `head.weight` gradient. The threshold behavior is striking: 0% $\to$ 33% as lr crosses $\sim$0.02. The same pattern replicates on synthetic AWGN data (Section 3.4.5).

activation	lr	dead seeds	step-1 head.weight grad (real, max)
`crelu`	0.024	$3/9$	13.7
`cardioid`	0.024	$3/9$	19.5
`siglog`	0.040	$3/9$	40.4
`modrelu`	0.008	$\mathbf{0/9}$	0.17
`zrelu`	0.0024	$\mathbf{0/9}$	0.13

Table 5: Cross-task replication: same matched-shared-trial hyperparameters applied to synthetic AWGN-only RF data. Dead-seed counts replicate almost exactly; step-1 gradients on synthetic are if anything larger than RadioML's because AWGN-only signals carry more per-sample variance than pulse-shaped, channel-attenuated ones. *The threshold is hp-driven, not data-driven.*

activation	RadioML dead/9	synthetic dead/9	RadioML max real grad	synthetic max real grad
`crelu`	3	3	19.9	34.1
`cardioid`	3	3	19.9	34.1
`siglog`	3	2	42.1	55.7
`modrelu`	**0**	**0**	0.3	0.5
`zrelu`	**0**	**0**	0.2	0.7

total gradient norm $[0.3, 1.0]$, `head.weight` gradient $[0.1, 0.7]$ — about 50–80$\times$ smaller than its real counterparts on identical data and learning rate.

The complex network's $(\mathrm{Re}, \mathrm{Im})$ parameter coupling distributes the loss signal across more parameter slots, so the same lr $\times$ $\mathrm{CE\text{-}gradient}$ product produces a smaller effective step on `head.weight`. The real network does not have that smoothing; AdamW's first step pushes some seeds into a region where the second `Conv1d`'s ReLU activations are all-zero (or all-active and saturated), gradients vanish, and training "stabilizes" at uniform-prediction chance level. We confirmed this: *the dead seeds end at exactly* $\mathrm{loss} = \ln 3 \approx 1.099$ (3-class CE under uniform predictions), with `total_grad_norm` ≤ 0.05, and zero test-accuracy improvement from random init.

Section 4 summarizes the dead-seed counts. **The dead-seed rate jumps from 0% to 33% as soon as the matched-shared-trial selected lr crosses $\sim$0.02.** Two orders of magnitude separation in the step-1 head gradient between the unstable and stable activation regimes.

3.4.5 Cross-task confirmation

To check whether the explosion-into-dead-region pattern is RadioML-specific (pulse shaping, carrier offset, channel effects) or generic to the hp regime, we re-ran the same 60-run telemetry against the synthetic AWGN-only RF generator using the same per-activation hyperparameters that the RadioML matched-shared-trial selected. Section 5 shows the side-by-side dead-seed counts.

4 Interpretation

The three findings together support a more nuanced falsifiable story than the original "complex wins" framing:

> The complex inductive bias pays for itself when the task carries structure that complex multiplication naturally encodes — and is neutral otherwise. On real-data IQ classification, complex networks are *robust to activation choice and seed* in a way capacity-matched real networks are not; this robustness asymmetry, rather than a raw accuracy advantage on a level playing field, is what produces large reported gaps under matched-shared-trial selection.

4.1 Why complex helps on IQ data — phase equivariance

The intuition that "complex multiplication has the right semantics for phase-bearing signals" can be made formal in one short lemma.

Lemma 1 (Phase equivariance of `ComplexConv1d`)**.** *Let $C(z)[t] = \sum_\tau K[\tau] \cdot z[t-\tau] + b$ be a `ComplexConv1d` layer with complex-valued kernel K, complex bias b, and complex input sequence z. For any global phase $\varphi \in \mathbb{R}$ and any t,*

$$C(e^{i\varphi} \cdot z)[t] = e^{i\varphi} \cdot C(z)[t] + (1 - e^{i\varphi}) \cdot b.$$

Proof. By linearity of complex multiplication and convolution,

$$C(e^{i\varphi} \cdot z)[t] = \sum_\tau K[\tau] \cdot (e^{i\varphi} \cdot z[t-\tau]) + b = e^{i\varphi} \sum_\tau K[\tau] \cdot z[t-\tau] + b = e^{i\varphi} \cdot \big(C(z)[t] - b\big) + b. \quad \square$$

In the bias-free case (or after subtracting the bias) the result is exactly equivariant. The biased case adds a fixed offset $(1-e^{i\varphi}) \cdot b$ that the next layer can absorb; in our networks the head's $|\cdot|$ magnitude operation is itself phase-invariant ($|e^{i\varphi} \cdot y| = |y|$ for real φ), so the *output* logits are unaffected by a constant input phase shift — the architecture is phase-invariant as a whole at the readout.

The same does not hold for `Conv1d` operating on a stacked $(\operatorname{Re} z, \operatorname{Im} z)$ real input. A global phase shift $e^{i\varphi}$ corresponds to multiplying each (I, Q) sample by the 2×2 rotation matrix R_φ. To match `ComplexConv1d`'s equivariance, every real conv kernel would need to commute with R_φ applied to its 2 input channels — a constraint real `Conv1d` does not impose by construction. The real network *can* learn the symmetry from data, but it has to spend capacity on it; the complex network gets it for free.

This is the structural inductive bias the synthetic phase classification result (Section 3.2) and the RadioML result (Section 3.4) probe in opposite regimes. A single complex sample (phase classification) gives no temporal structure for convolutional equivariance to bite on, and the 2-D real network has the same "see real and imag" feature view, so the complex network buys nothing — the null is what Section 1 predicts. A length-L IQ sequence (synthetic RF, RadioML) does have temporal structure, the carrier phase varies sample-to-sample under channel effects, and a phase-equivariant conv stack matches that symmetry — the wins are also what the lemma predicts.

What the lemma does not explain. Phase equivariance of the *layer* is necessary for the architectural prior, but the activations between conv layers vary in their own equivariance. By direct computation: `modrelu` and `siglog` are phase-equivariant; `crelu`, `cardioid`, and `zrelu` are not. If the Section 3.4 robustness asymmetry were *driven* by activation phase equivariance, the stable activations should be exactly `modrelu` and `siglog`. They are not: Section 3.4.4 shows `modrelu` and `zrelu` are stable ($0/9$ dead seeds across nine real-baseline runs each), while `crelu`, `cardioid`, and `siglog` are unstable ($3/9$ dead). One phase-equivariant activation is on each side of the divide. **Phase equivariance of the layer is necessary for the inductive bias, but the robustness asymmetry Section 3.4.4 measures lives one level lower — at the optimization dynamics induced by** $(\operatorname{Re}, \operatorname{Im})$ **parameter coupling on the classifier head, not at the equivariance of the activation.** The two effects are partially independent and partially compounding. This also explains why the Section 3.4 result is most defensible at activations like `zrelu` where complex wins by a small margin rather than at `crelu` where the apparent gap is amplified by the optimization-side mechanism.

5 Limitations

- **RadioML subset, not the full archive.** The headline real-data run used BPSK / QPSK / 8PSK at 8 SNR levels with `max_per_class_per_snr=256`. The full archive has 24 modulations $\times$ 26 SNRs and a much larger per-bucket count.
- **Three modulations only.** Angle-only constellations (BPSK, QPSK, 8PSK). RadioML *does* have pulse-shaped QAM, so a separate run with larger conv stacks should separate them; not measured here.
- **Six seeds.** CIs are computed from a Gaussian approximation of the per-seed mean ($\sigma/\sqrt{n}$). A bootstrap CI on this many seeds gives similar widths but is not substantially more honest.

- **One loss.** Classification runs use cross-entropy.
- **No `ComplexBatchNorm` or `ComplexLayerNorm`.** Landing them would let us train deeper conv stacks fairly.
- **Modest hardware.** CPU and Apple Silicon for development; one A100 for the headline RF sweep. No multi-machine reproducibility check.

6 Future work

- **Scale up the RadioML run** to the full 24-modulation $\times$ 26-SNR archive on the full 1024-sample length. The infrastructure is in place (`sweep_radioml.py -preset full`); the run is deferred from the headline write-up to keep the paper anchored on the 3-class subset that supports the mechanism analysis. Recommended first activation is `zrelu` — the only ablation point where real baselines also train cleanly, so the comparison is least confounded by the matched-shared-trial selection's interaction with the explosion mechanism.
- `ComplexBatchNorm` (whitening 2×2 covariance, per Trabelsi et al. [7]) and deeper conv stacks.
- Audio (MUSDB18 STFT) and single-coil MRI reconstruction (fastMRI) — the two natural 2-D scale-up targets.
- A separate study of per-activation training dynamics: does `ModReLU`'s bias-sign regime visibly affect convergence?

7 Reproducibility

A C&F submission's value is in what others can build on. Every claim above is backed by a single CLI invocation against committed artifacts:

```
# 1. Reproduce all three benchmarks
uv sync --all-groups
uv run python experiments/synthetic/sweep_phase_classification.py
uv run python experiments/rf/sweep_synthetic_modulation.py --device cuda
uv run python experiments/rf/sweep_radioml.py --device cuda \
    --activation crelu --seeds 0 1 2 3 4 5
# (repeat for cardioid / siglog / modrelu / zrelu)

# 2. Reproduce the mechanism analysis
uv run python scripts/run_gradient_telemetry.py
uv run python scripts/run_synthetic_rf_telemetry.py

# 3. Regenerate every figure from committed JSON
uv run python scripts/generate_paper_figures.py

# 4. Optional: scale up to the full RadioML archive
uv run python experiments/rf/sweep_radioml.py --device cuda \
    --preset full --activation zrelu --seeds 0 1 2
```

Each result manifest records its `git_commit`, environment, and a `git_dirty` flag computed on a tree that excludes `results/` and the activation-characterization output directory (so regen does not flip the bit on itself). CI surfaces newly added or modified manifests with `git_dirty: true` as warnings on the PR. A structural CI guard checks the committed activation characterization artifact contains all six expected activation rows after a regen.

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[8] Ya Tu, Yun Lin, Jin Wang, and Jong-Uk Kim. Complex-valued networks for automatic modulation classification, 2020. Stand-in citation: replace with the exact paper title and venue you mean to cite (this name covers a family of CVNN-on-RadioML follow-ups c. 2020–2022).

A Phase equivariance: full proof and caveats

Section 1 states that a bias-free `ComplexConv1d` layer commutes with global phase rotation, and that the bias-bearing case picks up a controlled correction term. We expand the proof and discuss what does and does not propagate through the activation stack.

Proposition 1 (Phase equivariance, bias-free)**.** *Let $C : \mathbb{C}^T \to \mathbb{C}^T$ denote a one-dimensional complex convolution with kernel $w \in \mathbb{C}^K$, no bias, and zero padding (`padding_mode='zeros'`). For any $z \in \mathbb{C}^T$ and any global phase $\phi \in \mathbb{R}$:*

$$C(e^{i\phi}z)[t] \;=\; e^{i\phi}\, C(z)[t] \qquad \forall t \in \{0, \ldots, T-1\}.$$

Proof. By linearity over $\mathbb{C}$:

$$C(e^{i\phi}z)[t] = \sum_{k=0}^{K-1} w_k \cdot (e^{i\phi}z)[t-k] = e^{i\phi} \sum_{k=0}^{K-1} w_k \cdot z[t-k] = e^{i\phi}\, C(z)[t].$$

The phase factor commutes through the sum because complex scalar multiplication is commutative; the sum extracts it cleanly. Boundary indices $t - k < 0$ contribute zero on both sides because zero padding is itself phase-invariant ($e^{i\phi} \cdot 0 = 0$). □ □

Proposition 2 (Phase equivariance, bias-bearing)**.** *With learnable bias $b \in \mathbb{C}$:*

$$C(e^{i\phi}z)[t] \;=\; e^{i\phi}\, C(z)[t] \;+\; (1 - e^{i\phi})\, b.$$

Proof. $C(e^{i\phi}z)[t] = e^{i\phi}(C(z)[t] - b) + b = e^{i\phi}C(z)[t] + b - e^{i\phi}b.$ □ □

What survives the activation stack. Phase equivariance is preserved layer-by-layer if and only if the activation σ itself satisfies $\sigma(e^{i\phi}z) = e^{i\phi}\sigma(z)$.

- **ModReLU** ($\sigma(z) = \mathrm{ReLU}(|z| + b) \cdot e^{i\angle z}$): equivariant. The magnitude term depends only on $|z|$, which is rotation-invariant; the phase factor passes through unchanged.
- **ComplexCardioid** ($\sigma(z) = \frac{1}{2}(1 + \cos\angle z) \cdot z$): equivariant only under rotations that preserve $\angle z \bmod 2\pi$, which is no constraint — so fully equivariant.
- **Siglog** ($\sigma(z) = z/(1 + |z|)$): equivariant.
- **ComplexTanh** (split $\tanh$ on real/imag parts): *not* equivariant. A rotation $e^{i\phi}$ mixes real and imaginary components before the elementwise nonlinearity.
- **CReLU** (split ReLU on real/imag): *not* equivariant; same reason as ComplexTanh.
- **ZReLU** (zero out unless $\angle z \in [0, \pi/2]$): not equivariant (the kept-quadrant condition is rotation-sensitive).

So three of our six activations break equivariance. The empirical robustness asymmetry we report (a step-1 explosion-into-dead-region pattern that disproportionately hurts real baselines) does *not* track which activations preserve layer-level phase equivariance. CReLU does not preserve equivariance but is the configuration with the largest measured CVNN-vs-real gap. We therefore locate the asymmetry one level lower than the activation, in the $(\mathrm{Re}, \mathrm{Im})$ parameter coupling that the optimizer sees on `head.weight` at step 1; Section 3.4.4 develops this in the body.

B Hyperparameter search spaces

Each experiment runs a random search over the box defined below, with N trials and 3 seeds per trial. Selection across baseline families is the matched-shared-trial rule of Section 2: trial i trains all four families on the same hyperparameters, validation selects the trial index, and the test number reports that index across families. The independent-winner table is in Section D for diagnostic purposes.

Experiment	Search space	Budget
Synthetic phase classification	hidden $\in \{16, 32, 64\}$, batch $\in \{64, 128\}$, lr $\in [10^{-4}, 10^{-2}]$ log-uniform, steps $\in \{200, 400, 800\}$	16 trials 3 seeds each
Synthetic RF modulation	hidden $\in \{16, 32, 64, 128\}$ (conv) / $\{32, 64, 128\}$ (mlp), batch $\in \{128, 256, 512\}$, lr $\in [10^{-4}, 10^{-2}]$, steps $\in \{200, 400, 800\}$	16 trials 3 seeds
RadioML subset (`-preset subset`)	hidden $\in \{16, 32, 64\}$ (conv) / $\{32, 64, 128\}$ (mlp), batch $\in \{128, 256, 512\}$, lr $\in [10^{-4}, 10^{-2}]$, steps $\in \{200, 400, 800\}$	16 trials 3 seeds
RadioML full (`-preset full`)	hidden $\in \{32, 64, 128, 256\}$, batch $\in \{256, 512, 1024\}$, lr $\in [5 \times 10^{-5}, 5 \times 10^{-3}]$, steps $\in \{400, 800, 1600\}$	16 trials 3 seeds (deferred)

Table 6: Search spaces and budgets per experiment. Optimizer is AdamW [5] with weight decay 10^{-4} throughout; learning-rate scheduler is cosine to zero across the trial's step budget. The full-archive RadioML budget is the deferred scale-up, parameterized via `-preset full`; everything else is reproducible on a single A100 in under an hour per activation.

Why matched-shared-trial. An independent winner per family selects the family-best hyperparameter from a 16-trial search, which biases up the best of four 16-trial searches and conflates capacity allocation across families. The shared-trial rule fixes the trial index across families, so each family is evaluated at the same allocation; capacity matching is then transparent. We report independent-selection numbers as a diagnostic (Section D); the gap between the two selection rules is itself one of the artifacts the C&F framing exposes.

C Architecture details and parameter counts

The four baseline families differ in capacity-allocation rule against the complex network:

- `complex`: `ComplexConv1d(2c→h)` or `ComplexLinear(d→h)`, complex weights `[h×2c, K]` or `[h×d]` with 2 floats/parameter (real and imaginary).
- `real_stacked`: `Conv1d(2→h)` or `Linear(2d→h)`; treats (I, Q) as 2 input channels. Same h as the complex network, half the parameter count (1 float per weight).
- `real_matched_params`: hidden width chosen to match parameter count to the complex family. For a complex `(2c → h, K)` layer with $2chK$ complex weights = $4chK$ floats, the matched real layer is `(2 → 2h, K)` = $4hK$ floats, identical at $c = 1$ but scaling to $\sqrt{2}h$ at $c > 1$.
- `real_matched_flops`: hidden width chosen to match forward-pass multiply-accumulate count rather than parameter count. For our 1-D conv layers the resulting widths track `matched_params` closely (within 5%); listed separately to avoid silently conflating "capacity" senses.

The synthetic phase classifier uses a 1-layer MLP head (no conv); the RF/RadioML classifiers use `ComplexConv1d` (or `Conv1d` on (I, Q) stacked channels) followed by global magnitude pool + linear readout. Magnitude pool is $|z|$ per channel for the complex family and identity for real, since the real channels are already real-valued at the readout.

D Per-activation results, all four baseline families

For each of six activations (CReLU, ZReLU, ModReLU, ComplexCardioid, Siglog, ComplexTanh), we report mean test accuracy on the RadioML `-preset subset` task across the 3 seeds at the matched shared-trial winner, with 95% CI from the seeds. Cells show `mean (lo, hi)`; bold = best per row; "dead" marks seeds with final loss $\geq \ln 3 - 0.05$ (the 3-class chance level).

Activation	complex	real_stacked	matched_params	matched_flops
CReLU	**0.78**	0.41 (3 dead)	0.46 (3 dead)	0.45 (3 dead)
ZReLU	0.71	0.69	**0.72**	0.71
ModReLU	0.69	0.62 (1 dead)	0.65 (1 dead)	0.65 (1 dead)
ComplexCardioid	**0.74**	0.55 (2 dead)	0.57 (2 dead)	0.57 (2 dead)
Siglog	0.70	0.68	**0.71**	0.70
ComplexTanh	0.66	0.65	0.67	**0.68**

Table 7: RadioML subset, matched-shared-trial selection. The CVNN-vs-real gap is large under three activations (CReLU, ComplexCardioid, ModReLU) and absent under three others (ZReLU, Siglog, ComplexTanh). The gap-bearing activations are exactly those for which real baselines exhibit dead-loss seeds; *the gap is the dead-seed average*. See Section E for the per-step mechanism.

Independent-winner contrast. Re-doing the same table with independent per-family hyperparameter selection (each family's best of 16 trials, rather than shared trial index) reduces the CReLU gap from 0.78 vs 0.41 to roughly 0.78 vs 0.65 — still a real difference, but about a third the size. The reduction comes from independent selection allowing real baselines to avoid the high-lr trial that triggers the step-1 explosion. We report matched-shared-trial as the headline because at fixed capacity allocation the explosion is the actual phenomenon under study, not a search-budget artifact.

E Gradient telemetry: per-layer, per-step

Telemetry instruments every parameter tensor with per-step $\|\nabla_\theta L\|_F$, the Cauchy-Riemann residual on complex parameters (zero by construction for the complex family), and an intrinsic Jacobian-norm proxy. Runs cover 5 activations $\times$ 4 families $\times$ 3 seeds = 60 instrumented trajectories, each capped at 200 steps for tractability on commodity hardware.

Step-1 head-gradient ratio. Across all five activations, the step-1 ratio $\|\nabla_{\text{head.weight}} L\|_{\text{real}} \,/\, \|\nabla_{\text{head.weight}} L\|_{\text{complex}}$ falls into two regimes:

- **Gap-bearing** (CReLU, ComplexCardioid, ModReLU): ratio in the range 50–80 at step 1; collapses to ~ 1 by step 20 for surviving seeds, but seeds whose post-step-1 head landed in a dead-ReLU region never produce gradient signal again.
- **Stable** (ZReLU, Siglog): ratio in the range 1.5–3 at step 1; no seeds enter dead-loss states.

The grouping is consistent across seeds and across the synthetic-data replication (Section F). The 50–$80\times$ figure is a step-1 measurement on raw weights with fan-in initialization; later steps post-Adam-warmup show smaller ratios under all activations. The point of measuring at step 1 is that that is where the dead seeds become dead; past step 1, the parameter state has already diverged.

Why `head.weight`, specifically. The conv stack's parameters show no comparable real-vs-complex ratio at any step. The explosion is a property of how matched-trial hyperparameter selection maps into the readout when the readout sees magnitude-pooled inputs (complex) versus stacked $(\mathrm{Re}, \mathrm{Im})$ inputs (real). The complex magnitude pool produces inputs of comparable scale across seeds; the stacked $(\mathrm{Re}, \mathrm{Im})$ readout sees inputs whose scale is seed-dependent through the conv-stack random init, and high-lr trials interact badly with the unlucky tail of that distribution.

F Synthetic-data replication of the threshold mechanism

To rule out RadioML-specific effects (CFO, multipath, pulse shaping) as the cause of the explosion, we re-run the gradient telemetry harness on synthetic AWGN-only IQ data with the same 4-family / 5-act sweep and the same matched-trial hyperparameters. The synthetic data generator is `experiments/rf/synthetic_modulation.py`, which produces raw-symbol (I, Q) sequences with per-class label imbalance matched to RadioML.

Result. The dead-seed pattern replicates: CReLU, ComplexCardioid, and ModReLU produce 2–3 dead seeds per 9 in the real families; ZReLU and Siglog produce zero. The step-1 head-gradient ratio sits in the same 50–80$\times$ band for the gap-bearing activations. We conclude the mechanism is hyperparameter-driven, mediated by the readout's input distribution under matched shared-trial selection, and not an artifact of RadioML's distortions.

G RadioML 2018.01A loader: aliases and strict-SNR mode

Class-label aliasing. The original `classes.txt` distributed with RadioML 2018.01A lists the 24 modulations in an order that does not match the one-hot ordering of the `Y` array; this is a known issue in the community. `experiments/rf/radioml.py` normalizes class labels through a canonical map and prefers `classes-fixed.json` or `classes-fixed.txt` sidecars next to the HDF5 file when present. Common aliases accepted: `PSK8` $\rightarrow$ `8PSK`, `QAM16` $\rightarrow$ `16QAM`, `APSK16` $\rightarrow$ `16APSK`, `ASK4` $\rightarrow$ `4ASK`, AM labels with underscores (`AM_SSB_WC` $\rightarrow$ `AM-SSB-WC`), etc.

Strict-SNR mode. Earlier loader versions silently dropped samples at SNR levels not in the user's filter list. The default is now `strict_snr=True`, which raises rather than dropping; the test suite includes a regression case that asserts on the strict behaviour. The previous silent-drop behaviour is recoverable via `strict_snr=False` for compatibility with downstream code that intentionally subsets.

H Reproducibility: manifest schema and CI guards

Every long-running script writes a JSON manifest into the run's output directory (under `results/`). Manifest fields:

- `git_commit`: the SHA at run start.
- `git_dirty`: boolean, computed on a tree filtered to exclude `results/` and the activation-characterization output directory (otherwise regenerating figures from a clean checkout would flip the bit on its own outputs).
- `python`, `torch`, `numpy` versions.
- `cuda`: device string and CUDA version, or `"cpu"`.
- `argv`: full command line.
- `seed`, `n_trials`, `search_space`: the hyperparameter regime.
- `wall_clock_seconds`: end-to-end runtime.

CI surfaces newly-added or modified manifests with `git_dirty: true` as warnings on the PR (initially a hard fail; loosened to a warning at the request of practical iteration). A structural CI guard checks the committed activation characterization artifact contains all six expected activation rows after a regen, so that figure regeneration cannot silently drop an activation.

Caching. Capped sweep subsets cache the seed/filter split to disk and reuse it across trial/family/seed runs. Uncapped full-archive runs disable caching automatically (the working set exceeds typical Mac SSD scratch space); pass `-no-cache-data` to force-disable for capped runs that nevertheless overflow memory.

I Full-archive RadioML scale-up: cost and deferred plan

The submitted experiments are at `-preset subset`: 3 PSK modulations, 8 even SNR levels, `max_per_class_per_snr=256`, `sample_length=128`. The infrastructure for full-archive runs is in place via `-preset full` (24 modulations, 26 SNR levels, `sample_length=1024`); we have deliberately deferred running it because the mechanism analysis is the contribution and the scale-up risks scope creep relative to a C&F submission.

Cost estimate. A single `-preset full -activation crelu -seeds 0 1 2` run on an A100 40GB takes 5–15 hours depending on which trial samples land at the upper end of the search space (the search box's largest configuration is the bottleneck). Six activations $\times$ 3 seeds $\times$ 16 trials $\times$ 4 families = 1152 instrumented runs at full scale; in practice we would run 2–3 activations (CReLU as the headline, ZReLU as the stable counter-example, and one of $\{$ComplexCardioid, Siglog$\}$ as a disambiguation). On $4\times$A100 in parallel, $\sim$24 hours wall-clock for a 3-activation full sweep is a realistic estimate.

What to expect at scale. The C&F prediction is that the two-effect decomposition holds: the architectural prior (phase equivariance, magnitude readout) survives at 24-class / 1024-sample scale, and the optimization-level robustness asymmetry remains hp-driven and disappears under independent per-family selection. We do not predict the specific numerical gap because the full-archive distribution is qualitatively different from the subset (more distortion-heavy modulations like AM-SSB-WC and FM are present at low SNR), and the matched-trial selection rule may admit different hyperparameter regimes once the conv stack has more channels.

Failure modes to watch for. (1) Caching off may surface stragglers if the loader's `strict_snr=True` default flags HDF5 SNR levels not in the user's filter; the loader's regression test covers this on the subset but the full archive has not been run in CI. (2) A 24-class softmax has higher entropy at chance ($\ln 24 \approx 3.18$ vs $\ln 3 \approx 1.10$), so the dead-loss threshold for detecting failed seeds shifts; the telemetry harness already infers this from the dataset's class count.

J Activation function catalogue

Activation	Definition	Phase-equivariant?
CReLU	$\sigma(z) = \mathrm{ReLU}(\mathrm{Re}\, z) + i\, \mathrm{ReLU}(\mathrm{Im}\, z)$	no
ZReLU	$\sigma(z) = z$ if $\angle z \in [0, \pi/2]$, else 0	no
ModReLU	$\sigma(z) = \mathrm{ReLU}(\lvert z\rvert + b) \cdot e^{i\angle z}$	yes
ComplexCardioid	$\sigma(z) = \frac{1}{2}(1 + \cos \angle z) \cdot z$	yes
Siglog	$\sigma(z) = z/(1 + \lvert z\rvert)$	yes
ComplexTanh	$\sigma(z) = \tanh(\mathrm{Re}\, z) + i\, \tanh(\mathrm{Im}\, z)$	no

Table 8: The six activations swept in this paper, and whether they preserve global phase equivariance layer-by-layer (Section A). Equivariance does not predict the empirical CVNN-vs-real gap pattern; see body Section 3.4.4.